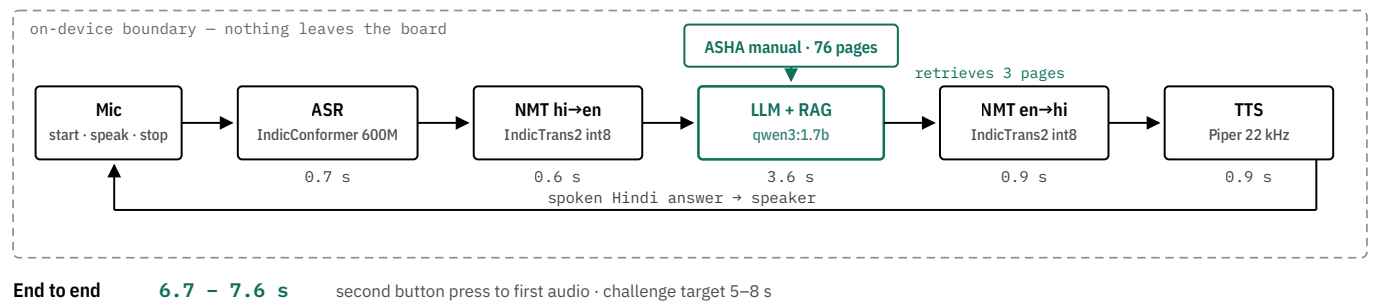


An ASHA worker presses one button, asks a question in Hindi, and hears an answer grounded in her own training manual — with no SIM, no Wi-Fi and no cloud call. Every figure on these two pages was measured on the device described above.

01 Pipeline



Hindi is understood and spoken at the edges; the reasoning core works in English, where the 1.7 B model is strongest. Retrieval sits between them, so the model answers only from the manual — when nothing matches, the language model is never called at all. Timings are measured from the second button press; ending on the silence timeout instead adds that 3.5 s window to the total.

02 Key design choices

CHOICE	WHY, AND WHAT IT COST OR SAVED	MEASURED
Text-only 1.7 B model, not a vision model	The shipping build does not use the camera, yet the vision model still loaded a CLIP tower and reserved a 322 MB vision compute buffer on every start. Its text-only sibling is the same family and reasoning behaviour.	3084 MB → 921 MB resident
Model chosen on safety, not on size	Smaller candidates were not merely weaker. With the rule “do not bathe a baby until 2000 gm” present verbatim in the retrieved page, a 1.5 B model answered “should be bathed daily until 2000 grams” — the negation inverted, on health guidance.	qwen2.5:1.5b inverted the rule; qwen3:1.7b answered correctly
Retrieval before the model, and refusal without it	Answers are grounded in the manual page, not model memory. If retrieval matches nothing the device says so instead of inventing — an unaided model happily answered “capital of France” and offered engine-repair advice.	Out-of-scope questions refused, no LLM call made
Long speech transcribed in chunks	The conformer was trained on short utterances and loses coherence over long ones, degrading into nonsense rather than stopping. Audio is split near 12 s at the quietest point so seams fall in word gaps.	38 s clip: 42% → 95% of words
Prompt cache capped	Every RAG prompt is unique, so llama.cpp’s 8 GB-limit prompt cache never hit and only grew — the real cause of the swap thrashing that stalled the board, not model size.	60 queries: +592 MB → +3 MB swap

03 Accuracy testing

Answers were being judged one question at a time, which cannot tell an improvement from a trade — a retrieval change that helped scenario questions had already been seen to break factual ones. So correctness is measured, not argued.

A fixed set of ten questions is drawn from the manual. Each carries the fact its answer must contain (2000 for the bathing rule, gauze for the airway step), plus one deliberately out-of-

PATH UNDER TEST	SCORE	FAILURE MODE
Text in → answer retrieval, model, prompt	10 / 10 —	

scope question that must be refused. Any change is re-scored in about a minute.

The gap between the two rows is entirely speech and translation noise: when the question reaches the reasoning core intact, the answer is right every time. That is what directed the remaining engineering effort at the input path rather than at the prompt.

PATH UNDER TEST	SCORE	FAILURE MODE
Speech in → answer full pipeline	8 / 10	Mis-heard word, or a trailing clause lost in translation
The harness found two real defects the eye had missed: a retrieval stemmer under which the question’s weighed could never match the manual’s weigh — making the one page that answers it unreachable — and an over-narrow test assertion that had been failing a correct answer.		

04 Latency

STAGE	TYPICAL	WHAT SETS IT, AND WHAT WAS DONE ABOUT IT
Speech → text	0.7 s	First question once cost 6–8 s: the warm-up exercised the ONNX sessions but not librosa’s JIT. Warming the real inference path moved that cost to boot.
Translate hi→en	0.6 s	Two direction models resident; the third (indic↔indic, 326 MB) is lazy-loaded and never touched, as every route goes through English.
Retrieve + answer	3.6 s	The budget’s largest item. Capped at 90 new tokens after a rambling answer once burned 17 s on its own; retrieval itself is under 10 ms.
Translate en→hi	0.9 s	Decode limit raised 64 → 192 tokens; at 64 a spoken answer stopped mid-instruction, dropping the verb from “requires immediate intervention”.
Text → speech	0.9 s	Piper at 22 kHz. Numbers are spelled out before synthesis — a bare “20” is not reliably voiced as बीस, and dosages and weeks are what these answers turn on.

05 Power efficiency

3.4 W IDLE, MODELS RESIDENT Board stays ready; nothing reloads between questions.	5.1 W AVERAGE DURING A QUERY Peak 9.7 W for roughly one second at generation.	15 W ENFORCED POWER MODE Chosen over 25 W: the higher cap browned the board out under load.
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Memory is the binding constraint on this board, and power follows it. Steady state is 4.2 GB of 7.4 GB — OS 1715 MB, language model 921 MB, speech and translation service 1593 MB — leaving headroom that stays flat across sustained use: 60 consecutive queries moved swap by 3 MB.

06 Engineering quality & code

How the work is structured

- Two repositories, 23 reviewed commits, each recording the measurement that motivated it — a later reader can tell why a constant is 3.5 and not 2.0.
- Four systemd units in dependency order. Load order matters: the language model must claim GPU memory before the speech service, or it silently splits across CPU and runs many times slower.
- Every model runs from local disk. The device is airplane-mode capable by construction, not by configuration.
- Superseded models are kept in place so a voice or checkpoint can be rolled back without a re-download.

Failures designed for

- Recording is ended explicitly. The trigger pin cannot report a sustained press — it drops after 7.8 s while the button is still held — so hold-to-talk was cutting questions mid-sentence, losing a नहीं and inverting the answer with it. One press now starts the recording and the next ends it, with a 3.5 s silence timeout as fallback.
- Empty or unintelligible speech is answered with a spoken Hindi prompt to repeat, never with a guess.
- Every stage that calls another service retries and reports, so one slow subsystem cannot wedge the interaction loop.
- Freed memory is returned to the OS after each interaction; on a board with no spare RAM, “free it later” is the same as leaking it.